

AI Face Attendance: Daily Operations & Verification Guide

1. Daily Operations Lifecycle (Start se Lekar Stop Tak)

Phase	Command	Purpose & Explanation
1. Folder	<code>cd ~/Music/AI_Face_recognition/backend</code>	Project directory me enter karne ke liye.
2. Venv	<code>source venv/bin/activate</code>	Python virtual environment activate karta hai.
3. Containers	<code>docker compose up -d</code>	MySQL (3306) aur Typesense (8109) background me run karta hai.
4. Run App	<code>uvicorn main:app --host 0.0.0.0 --port 8000 --reload</code>	AI Face detector, webcam loop aur FastAPI server chalu karta hai.
5. Stop App	<code>Ctrl + C (Terminal) / 'q' (Camera)</code>	Camera hardware release karke Python server stop karta hai.
6. Stop Docker	<code>docker compose down</code>	Containers safely stop karke RAM aur ports free karta hai.
7. Exit Venv	<code>deactivate</code>	Virtual environment close karne ke liye.

2. Docker Database & Tables Check Commands

Target	Command	Explanation
Direct MySQL	<code>docker exec -it attendance_mysql mysql -u ai_attendance -p2003 -e 'SHOW DATABASES; USE Ai_attendance; SHOW TABLES;'</code>	Confirm karta hai ki Ai_attendance DB aur attendance_logs table bani hai.
Interactive Shell	<code>docker exec -it attendance_mysql mysql -u ai_attendance -p2003</code>	Container ke andar MySQL console (mysql>) open karta hai manual queries ke liye.
Table Schema	<code>docker exec -it attendance_mysql mysql -u ai_attendance -p2003 -e 'DESCRIBE Ai_attendance.attendance_logs;'</code>	Columns: id, user_id, name, login_time, logout_time, duration verify karta hai.
Live Attendance	<code>docker exec -it attendance_mysql mysql -u ai_attendance -p2003 -e 'SELECT * FROM Ai_attendance.attendance_logs;'</code>	Camera dwara dynamically mark kiye gaye live login/logout records dikhata hai.
Typesense Vector	<code>curl -H 'X-Typesense-API-KEY: xyz123' http://localhost:8109/collections</code>	face_embeddings collection aur 512-dim vectors verify karta hai.

3. Daily One-Liner Shortcuts

Fast Start: `cd ~/Music/AI_Face_recognition/backend && source venv/bin/activate && docker compose up -d && uvicorn main:app --host 0.0.0.0 --port 8000 --reload`

Safe Stop: Terminal me Ctrl+C dabayein, fir: `docker compose down && deactivate`

Browser Check: <http://localhost:8000/attendance/logs>